

NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2 PRELIMINARY EXAMINATION
Higher 3

MATHEMATICS

9824/01

Paper 1

16 September 2016

3 hours

Additional Materials: Answer Paper
 List of Formulae (MF15)
 Cover Sheet

READ THESE INSTRUCTIONS FIRST

Write your name, registration number, subject tutorial group, on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **6** printed pages.



National Junior College

[Turn over

- 1 In this question, you may assume that if $k_1, \dots, k_n$ are distinct positive real numbers, then

$$\frac{1}{n} \sum_{r=1}^n k_r > \left(\prod_{r=1}^n k_r \right)^{\frac{1}{n}},$$

i.e. their arithmetic mean is greater than their geometric mean.

Suppose that a, b, c and d are positive real numbers such that the polynomial

$$f(x) = x^4 - 4ax^3 + 6b^2x^2 - 4c^3x + d^4$$

has four distinct positive roots.

- (i) By considering the relationship between the coefficients of f and its roots, show that $c > d$. [4]
- (ii) By differentiating f , show that $b > c$. [3]

- 2 The Fibonacci numbers F_n are defined by the conditions $F_0 = 0$, $F_1 = 1$ and

$$F_{n+1} = F_n + F_{n-1}, \text{ for all } n \geq 1.$$

- (i) Compute F_2, F_3, F_4 and F_5 . [2]
- (ii) Compute $F_{n+1}F_{n-1} - F_n^2$ for a few values of n and give a conjecture for its general formula. Prove your conjecture by induction, or otherwise. [5]
- (iii) By induction on k , or otherwise, show that $F_{n+k} = F_k F_{n+1} + F_{k-1} F_n$ for all positive integers n and k . [4]

- 3 (i) Show that $\frac{1 + \cos \theta}{\sin \theta} = \tan\left(\frac{\pi}{2} - \frac{\theta}{2}\right)$, where $t = \tan\left(\frac{\theta}{2}\right)$. [2]

- (ii) Using the substitution $t = \tan\left(\frac{\theta}{2}\right)$ to show that, for $0 < \alpha < \frac{\pi}{2}$

$$I = \int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos \alpha \sin \theta} d\theta = \int_0^1 \frac{2}{(t + \cos \alpha)^2 + \sin^2 \alpha} dt. \quad [3]$$

- (iii) By means of the further substitution $t + \cos \alpha = \sin \alpha \tan u$, show that

$$I = \frac{\alpha}{\sin \alpha}. \quad [3]$$

- (iv) Deduce a similar result for $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin \alpha \cos \theta} d\theta$. [3]

- 4 Use the binomial theorem to expand $\left(1 + \frac{x}{n}\right)^n$, where n is a positive integer, writing your expansion in the form

$$1 + x + a_2 \left(\frac{x^2}{2!}\right) + a_3 \left(\frac{x^3}{3!}\right) + \dots + a_n \left(\frac{x^n}{n!}\right).$$

Express a_2 and a_3 explicitly in terms of n , and show that

$$a_r = \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{r-1}{n}\right) \text{ for } 2 \leq r \leq n. \quad [5]$$

Give a simplified expression for a_n . [2]

Show that, for $2 \leq r \leq n$,

$$(i) \quad 0 < a_r < 1, \quad [2]$$

$$(ii) \quad \frac{1}{r!} \leq \frac{1}{2^{r-1}}. \quad [2]$$

Hence show that, when x is positive,

$$\left(1 + \frac{x}{n}\right)^n < 1 + x + \frac{x^2}{2} + \frac{x^3}{2^2} + \dots + \frac{x^n}{2^{n-1}}. \quad [2]$$

Assuming now that $0 < x < 2$, by summing the geometrical progression that appears after the first term on the right hand side, show that $\left(1 + \frac{x}{n}\right)^n < \frac{2+x}{2-x}$, for all positive integer values of n . [3]

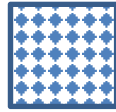
- 5(a) (i) Let $S(r, n)$ denote the number of ways of distributing r distinct objects into n identical boxes so that no box is empty.

Prove by induction that $S(r, 3) = \frac{1}{2}(3^{r-1} - 2^r + 1)$, for $r \geq 3$. [5]

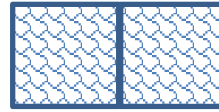
- (ii) Find the number of ways such that Mrs Wang can distribute 10 different shirts into 3 distinct drawers such that there is no empty drawer. [2]

- (b) A $1 \times n$ rectangular wall is to be fully paved by two tiles type of different sizes which are 1×1 and 1×2 as shown in the diagram below.

Tile Type 1



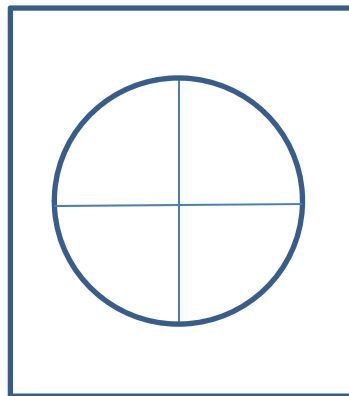
Tile Type 2



Find the number of ways to pave the 1×4 rectangular wall either with tile type 1 or tile type 2 or both. [1]

Suppose there are additional tile type of 1×3 and another tile type of 1×2 of different design to select. Let a_n be the number of ways to pave the $1 \times n$ rectangular wall. Find a recurrence relation for a_n . State the initial conditions. [4]

- 6 A bag designer wants to design a tote bag such that there is a circle printed on the front which is divided into four congruent quadrants as shown in the diagram below. Each of the four quadrant is printed with one of the 10 distinguishable colours.



- (i) How many different designs of circles are there
- (a) in total, [1]
 - (b) with reflection symmetry in a horizontal line, [1]
 - (c) with no rotational symmetry and no reflection symmetry? [3]
- (ii) Suppose there are c colours and all c colours are distinguishable, where $c \geq 4$. The bag designer decides that there should be no pair of bags such that the circle on one of the tote bag is a reflection of, or a 180° rotation of, the circle on the other tote bag. Show that the maximum number of such bags that the bag designer can design is

$$\frac{1}{4}c^2(c^2 + 3). \quad [3]$$

Terry chooses 5 tote bags from a range of tote bags of different designs and decides to give a tote bag to each of his 8 friends. Find the number of ways that he can do this, buying at least one of each of the 5 designs. [5]

7(a) It is given that $y = f(x)$ is a particular solution of the differential equation

$$\frac{dy}{dx} = y(x + y)$$

and $f(0) = 1$.

Use the improved Euler method with step size 0.5 to estimate y at $x = 1$.

[3]

(b) (i) Show that the substitution $z = x \frac{dy}{dx} + y$ reduces the differential equation

$$x(1 + x^2) \frac{d^2y}{dx^2} + 2(1 + 2x^2) \frac{dy}{dx} + x(2y - 1) = 0$$

to

$$\frac{dz}{dx} + \frac{2x}{1 + x^2} z = \frac{x}{1 + x^2}.$$

Hence, find y in terms of x .

[9]

(ii) It is given that

$$y = g(x)$$

is a solution to the differential equation such that its graph has an axis of symmetry. On a single diagram, sketch three possible graphs of $y = g(x)$, such that no two of the three graphs can be obtained from each other via linear transformation(s).

[3]

8 A body with mass 5 kg is attached both to a horizontally suspended spring and to a dashpot with a damping constant k Ns/m. It is known that the body stretches the same spring 0.4 m when it is suspended vertically. The body is now pulled 0.2 m to the right, stretching the spring, and set in motion with an initial velocity of 3 m/s.

[You may assume that the gravitational acceleration is 9.8 m/s^2]

(i) Show that the motion of the body can be modelled by the equation

$$10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 245x = 0,$$

where x is the body's displacement from its equilibrium position at time t .

[2]

(ii) Given that $k = 40$, find an expression for x in terms of t .

[4]

- 9 A fishery cultivates wrasse fish to satisfy the increasing consumption of fish as nutritious diet by its people. The following logistic growth equation is used to model the wrasse fish population, W , in the fishery, after t months in the absence of fishing:

$$\frac{dW}{dt} = rW \left(1 - \frac{W}{M} \right)$$

- (i) Explain the significance of the constants r and M in this model. [2]

The fishery's surface area is 800m^2 and 1m^2 of it can sustain 20 wrasse fish. A number of licenses, h , is issued per month to fishermen to harvest the wrasse fish. Past data has shown that each fisherman with a license can harvest an average of 200 fish per month.

- (ii) Write down, in terms of r , the modified logistic growth equation, taking into consideration the harvesting of the wrasse fish. [1]

- (iii) Given that $r = 0.5$, sketch a bifurcation diagram for the parameter introduced in (ii), indicating clearly its bifurcation value and the equilibrium points when there is no harvesting. [4]

- (iv) Suppose at the beginning, the fishery decides to issue only 10 licenses per month to the fishermen. What is the least initial number of wrasse fish required to sustain harvesting in the long term? [2]

– END OF PAPER –